

Bethany (Beijia) Yuan

by7175@princeton.edu | 858-531-8630 | <https://beijia-yuan.github.io/byuan.github.io/>

EDUCATION

Princeton University

Aug 2024 – Present

Science GPA: 4.00, overall GPA: 3.83

Biophysics, PhD candidate

Advisors: William Jacobs, Gautam Reddy

University of California, San Diego

Sep 2019 – Jun 2023

Summa Cum Laude, GPA: 3.99

Bioengineering: Bioinformatics, Bachelor of Science

General Physics, Bachelor of Arts

PUBLICATION

Z. Liu*, **B. Yuan***, M. Rao, G. Reddy, W. Jacobs. A thermodynamic metric quantitatively predicts disordered protein partitioning and multicomponent phase behavior. *In Review with preprint available*.

S.M. Epp, G. Castrillón, **B. Yuan**, J. Andrews-Hanna, C. Preibisch, V. Riedl. BOLD signal changes can oppose oxygen metabolism across the human cortex. *Nat Neurosci* (2025).

RESEARCH EXPERIENCE

Decoding the Molecular Determinants of Intrinsically Disordered Protein Phase Behaviors

Jacobs Group at Princeton Chemistry – collaborated with Zhuang Liu

Dec 2024 – Present

- Built an encoder-decoder machine learning-based framework to find IDP sequence features determining its phase behaviors and predict IDP mixture phase behaviors.
- Designed the model architecture to preserve all symmetries of a valid free energy function.
- Trained on MD simulation data, the model quantitatively predicted context-dependent IDP phase behaviors and unified prior classifier-based phase separation predictors.
- Developed a geometric interpretation for the low-dimensional feature space found by the model.

Inferring Multi-environment Genotype-Fitness Mapping with High-throughput QTL in Yeast

Reddy Group at Princeton Physics

Mar 2025 – Present

- Built a machine learning-based framework to find causal gene loci and model their interactions in contributing to yeast fitness across environments.
- Explored combinations of different architectures including transformer, MLP and linear model to probe different hypotheses in the nature of the gene interactions.
- Planned to investigate the model's ability to generalize to new genotypes and new environments with few-shot learning

Spatiotemporal Organization of the Genome in Cell Nucleus

Dudko Group at UCSD Physics

Jul 2022 – Aug 2024

- Modeled chromatin dynamics as a self-avoiding random walk to characterize site-site interactions in key molecular processes, including enhancer-promoter interactions.
- Developed an analytical expression to approximate the end-to-end distance probability distribution between two interaction sites by introducing radius-of-gyration dependence into the Flory model.

- The model provided explanation for loops both facilitating and insulating chromatin site-site interactions.
- Designed molecular dynamics simulations and Monte Carlo simulations for model validation.

Zinc Finger Induced Reprogramming of Endothelial Cells under Shear Stress

Subramaniam and Chien Lab at UCSD Bioengineering – collaborated with Dyvia Murali Dec 2019 – Aug 2024

- Investigated the role of zinc finger proteins in endothelial cell response to shear stress.
- Led the analysis of time-series bulk and single cell RNA-seq, siRNA microarray, and ChIP-seq data.
- Curated a comprehensive zinc finger protein database.
- Independently designed and pitched perturbation experiments to a collaborating lab, then conducted the approved studies, including siRNA transfection, microfluidic shear stress application, and qPCR.

Quantitative Blood-Oxygen-Level-Dependent (BOLD) Signaling in MR Imaging

Riedl Lab at TU Munich, Germany

Jul 2022 – Sep 2022

- Independently formulated an empirical derivation for the Davis Model, pinpointing the assumptions.
- Selected the optimal brain activity quantification method - combining Fujita's approach with BOLD signal measurements - by performing linear regression on cerebral metabolic rate of oxygen (CMRO2) data.
- Showcased findings at the Practical Research Experience Program poster session.

Multi-Omics Modeling of Glucose Metabolism

Kuroda Lab at University of Tokyo, Japan

Sep 2021 – Nov 2021

- Integrated kinetics differential equations with Western blot data for the Central metabolic pathway.
- Delivered a presentation for the Research Internship (UTRIP) Program.

TEACHING

Teaching assistant for Integrated Science Curriculum (ISC) at Princeton

Aug 2025 – Dec 2025

Designed and taught precepts together with the teaching team

Prepared homework solutions and graded homework and exams

Tutor for Computer Science and Engineering at UCSD

Designed and graded homework and exams and held weekly office hours.

CSE21: Mathematics for Algorithms and Systems Analysis

Apr 2021 – Jun 2021

CSE181: Molecular Sequence Analysis

Jan 2022 – Mar 2022

Math, Physics, and CS Academic Coach for PATHS Program at UCSD

Sep 2021 – Jun 2023

Held 15h of weekly office hours and assisted in program events to support underrepresented STEM students.

MATH 3,4,10 and 20 series: Algebra and Calculus

PHYS 1, 2 series and lab: Calculus based Mechanics, EM, and Waves, Optics and Modern Physics

CSE 8A, 8B, 11: Introduction to Programming and Computational Problem-Solving

SCIENTIFIC OUTREACH

Content Creator for The BigBENG at UCSD

Sep 2020 – Jun 2022

- Wrote scripts for Khan-Academy like videos for difficult bioengineering courses, uploaded on YouTube.

Leader of Translational Medicine Day Committee, BMES at UCSD

Sep 2019 – Jun 2022

- Interviewed professionals and created educational videos to raise the awareness in translational medicine.